

Algorithms and Analysis

What this course will cover

Optimization:

$x \in \mathbb{R}^n$, ($n=2$ in this lecture)

$$\begin{array}{ll} \text{minimize :} & f(x) \\ \text{subject to :} & x \in \mathcal{X}. \end{array}$$

What are the basic algorithms for solving these convex problems? How do we develop new algorithms? How fast do they run, and what computing resources do they need?

Examples of problems this course will cover

M-Estimators —

[Least Squares Regression
Ridge (L2-regularized) Regression
Lasso (L1-regularized Regression)
Logistic Regression

General convex functions for which we can compute derivatives.

Ridge Regression

Given data: $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

want to find a linear relationship between x and y : $y \approx x^T \beta$

$$\min: \frac{1}{n} \sum_{i=1}^n (x_i^T \beta - y_i)^2 + \lambda \sum_{j=1}^d \beta_j^2$$

more compactly:

$$\min: \frac{1}{n} \|X\beta - y\|_2^2 + \lambda \|\beta\|_2^2$$

$$\text{st: } \beta \in \mathbb{R}^d$$

Ridge Regression

$$f(\beta) = \frac{1}{n} \|X\beta - y\|_2^2 + \lambda \|\beta\|_2^2$$

Gradient descent

Initialize: β_0

GD update:

$$\beta_{t+1} = \beta_t - \eta \nabla f(\beta_t)$$

Annotations for the GD update:

- β_t : where we are after t iterations
- η : step size
- $\nabla f(\beta_t)$: gradient of f

Newton's method

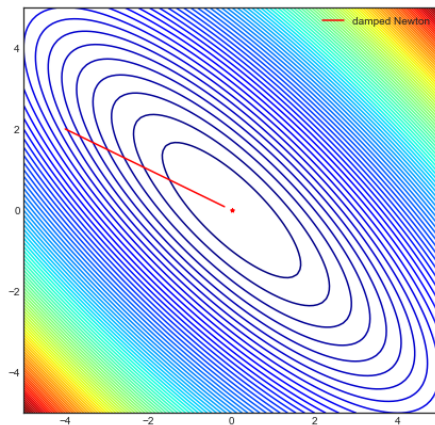
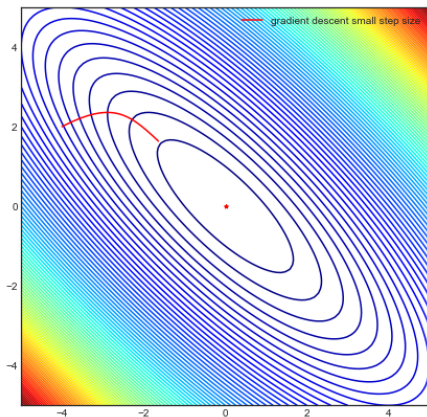
Initialize: β_0

Newton update:

$$\beta_{t+1} = \beta_t - \underbrace{\eta (\nabla^2 f(\beta_t))^{-1}}_{\text{inverse Hessian}} \nabla f(\beta_t)$$

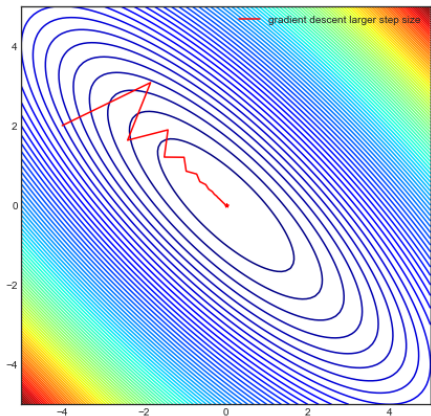
Ridge Regression

Very small step size η

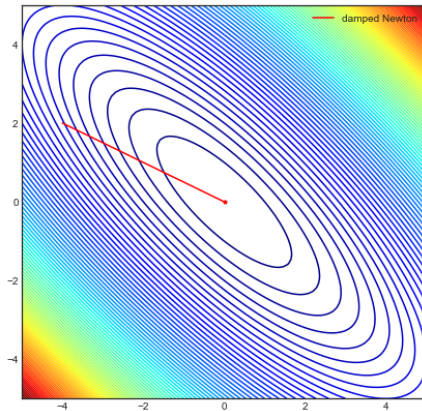


Ridge Regression

Larger step size η



Gradient with
larger step size

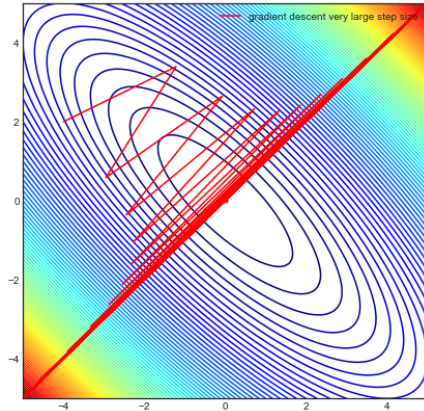


Newton

Ridge Regression

Gradient descent with much larger (too large)

step size.



Fails to converge
- diverges

- ① What accounts for difference between GD and Newton?
- ② Why would anyone use GD when we have Newton?
- ③ What is the correct/best step size?

Lasso — L1 regularized regression

$$\text{Data: } \{(x_1, y_1), \dots, (x_n, y_n)\} \quad X^T \beta \approx y$$

$$\min: \frac{1}{n} \sum_{i=1}^n (x_i^T \beta - y_i)^2 + \lambda \sum |\beta_i|$$

more compactly

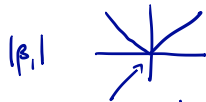
$$\min: \frac{1}{n} \|X\beta - y\|_2^2 + \lambda \|\beta\|_1$$

← L1 norm

This is a very useful & powerful formulation when we believe that β is sparse or nearly sparse — i.e., many entries of β are zero or close to zero.

Lasso — L1 regularized regression

$$\min: \frac{1}{n} \|X\beta - y\|_2^2 + \lambda \underbrace{\|\beta\|_1}$$



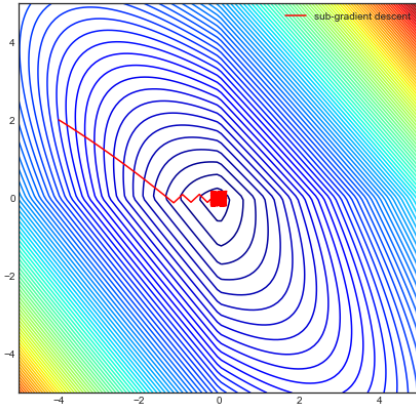
not differentiable
also, no second derivative

sub gradient method / proximal gradient method $\Rightarrow$ no Hessian

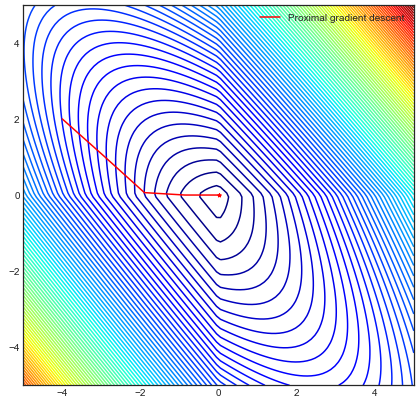
very similar in
spirit to GD

Lasso — L1 regularized regression

sub-gradient method



proximal gradient method



- ① Why does sub-grad. method cycle at the end? Can we fix that?
- ② What is this prox-grad method, and where can I learn more...?



In Summary: This class —

Fundamental (broad) problem classes

Algorithms tailored for these

Analysis of convergence rates

Implications for Machine Learning problems

This will include all the examples we saw above: gradient descent, subgradient method, Newton's method, proximal gradient method.